ANNEXURE-II

Technical Evaluation & Compliance Questionnaire

Challenge Reference: iDEX Drishti Challenge – AI Tool for Design Optimization (Class Rules, Standards & Regulations Compliance) & 3D–BOM Synchronization (Sr. No.71)

Issuing Authority: MDL

Stage: Technical Scrutiny & Shortlisting (Post Eligibility Verification)

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1. Purpose & How to Use This Document

1.1 Purpose

The purpose of this questionnaire is to enable a **structured, objective and uniform technical evaluation** of bidders who have cleared the initial eligibility stage (Startup/MSME status and relevant experience with supporting Work Orders / Work Done Certificates).

The information furnished will be used to assess each bidder's **technical capability, solution maturity, domain relevance, security posture, deployment readiness and ability to execute** the proposed solution within a Defence Shipyard environment. The objective is to identify genuinely capable vendors and shortlist a focused set for the subsequent stage.

1.2 Nature of this Stage

- This is a **capability and compliance assessment**, not a request to build a free custom solution.
- Where a demonstration is sought (Section 13), it may be satisfied using the bidder's **existing product, prior implementations, or a small representative sample** , the intent is to assess technical maturity, not to mandate bespoke development at this stage.
- All bidders are evaluated against the **same published criteria and scoring rubric** (Section 15).

1.3 Instructions to Bidders

1. Answer **every** question. Where a question is not applicable, state "Not Applicable" with a brief reason. Blank responses will be scored as "not demonstrated".
2. Provide **descriptive answers** with specifics (tools, versions, metrics, references). Avoid unsubstantiated marketing claims.
3. Support every claim of experience with **documentary evidence** (Work Orders, Completion/Work Done Certificates, client references, screenshots, demo links).
4. Keep the **Technical Solution Summary (Section 5–6) within 10 pages**, excluding annexures and diagrams.
5. Submit responses in the **prescribed tables** wherever provided. Do not alter column headings.
6. Furnish a signed **Declaration of Correctness** (Annexure A). Misrepresentation may lead to disqualification at any stage.

1.4 Confidentiality & Intellectual Property

- Information submitted will be used solely for evaluation under this challenge and treated as commercially confidential.
- Ownership of IPR will be held jointly by MDL and Bidder.

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2. DRISHTI Problem Statement:

Develop an AI-powered system that analyses detail design parameters with respect to class rules, standards and regulations, predicts downstream impacts across disciplines (hull, propulsion, weapons and concerned systems, including structural re-analysis/weight redistribution and re-qualification/trials), and provides intelligent inputs towards finalisation of detail design, incorporation of the design into basic drawings before submission to class, and incorporation of the latest approved revision in the production drawings post approval of the class. The system shall additionally synchronise 3D models with the Bill of Material (BOM) to eliminate human errors.

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3. Bidder Information & Eligibility Reconfirmation

#	Field	Response
3.1	Company / Entity Name	Arcaisys Pvt Ltd
3.2	Startup / MSME status	Startup
3.3	DPIIT Recognition No. (if Startup)	DIPP261576
3.4	Udyam Registration No. (if MSME)	-
3.5	Year of Establishment	2026
3.6	Registered / Head Office (City, State)	Mumbai, Maharashtra
3.7	Development Centre location(s) (India / Overseas)	India
3.8	GST No. / PAN	ABFCA5830P
3.9	Ultimate beneficial ownership , any foreign / land-border-country holding? (Yes/No; if Yes, details)	No
3.10	Website	https://arcaisys.com
3.11	Single Point of Contact (Name, Designation, Email, Phone)	Mr. Leejo Purvathan, Director, leejo.purvathan@arcaisys.com, +91 9029767757

3.12 Team Composition

Role	No. of Full-Time Personnel	Avg. Relevant Experience (Years)
AI / ML Engineers	7	6
Computer Vision Engineers	4	5
3D / Point Cloud / Geospatial Engineers	1	2

CAD / CAE Engineers	1	20+
Mechanical / Marine / Piping Engineers	3	10
Shipbuilding / Naval Domain Experts	3	35+
DevOps / Deployment / Security Engineers	4	5
Total Technical Headcount	23	-

3.13 Certifications (attach copies)

Certification	Held? (Y/N)	Reference / Validity
ISO 9001 (Quality)	N	-
ISO/IEC 27001 (Information Security)	N	-
CMMI Level (specify)	N	-
CERT-In empanelment / other security certification	N	-
Other relevant certifications	IBM Machine Learning	https://courses.yl-ptech.skillsnetwork.site/certificates/7f361d46c5a74a5f8d0be30e0979c514

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4. Relevant & Similar Project Experience

4A. Relevant Domain Experience , repeat the block for each project (max 8 projects)

Field	Response
Client / End-User	IRCLASS Systems and Solutions Private Limited.

Sector (Engineering / Manufacturing / Shipbuilding / Defence / Govt / PSU / Technical KM / Other)	Ship Auditing Software
Project Title & Brief Scope	IRS Auditing Software for ships which can be taken in pendrive and installed in ship machine.
Duration (From – To)	20th March 2013 – 15th Oct 2013
Order Value (₹)	-
AI/GenAI components used (CV / OCR / DL / LLM / RAG / Knowledge Graph / Object Detection / 3D)	** Worked as a developer for this software
Deployment environment (Cloud / On-Prem / Air-Gapped)	On-Prem
Current status (Live / Completed / Ongoing)	Completed
Evidence attached (Work Order ☐ Completion/Work Done Certificate ☐ Client Reference ☐)	Work Experience letter can be shown whenever required.

Field	**Response**
Client / End-User	Internal
Sector (Engineering / Manufacturing / Shipbuilding / Defence / Govt / PSU / Technical KM / Other)	Software Engineering
Project Title & Brief Scope	Meridian Lite. Ai hallucination Reduction and Prevention Layer
Duration (From – To)	15th May – Till date (Proposed end date by December)
Order Value (₹)	-
AI/GenAI components used (CV / OCR / DL / LLM / RAG / Knowledge Graph / Object Detection / 3D)	Knowledge Graph, LLM, RAG system
Deployment environment (Cloud / On-Prem / Air-Gapped)	Cloud
Current status (Live / Completed / Ongoing)	Ongoing
Evidence attached (Work Order ☐ Completion/Work Done Certificate ☐ Client Reference ☐)	-

4B. Similar / Closely-Aligned Experience , repeat for each project (max 5 projects)

Cite projects that involved the topics that are related to the problem.

Field	Response
Client / End-User	Synergy Medtech Solutions
Problem solved (map to this challenge)	Developed an AI-powered engineering decision support platform that automated document interpretation, regulatory compliance verification, dependency analysis, impact assessment, and workflow-based approvals. The solution addressed complex interdependent data, enabling traceable decision-making, rule validation, change impact analysis, and intelligent recommendations,capabilities directly aligned with the MDL requirement for AI-assisted design compliance, downstream impact prediction, and engineering workflow automation.
Input data types, formats and sources handled as a part of similar projects.	Structured and unstructured enterprise data including technical specifications, regulatory and standards documents, PDFs, Word documents (DOC/DOCX), spreadsheets (XLS/XLSX/CSV), databases, transactional records, APIs, scanned documents (OCR), emails, master data, workflow records, and business system integrations. Similar to MDL, the platform consolidated heterogeneous data into a unified searchable knowledge base for AI-assisted analysis.
AI/CV techniques applied	Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Named Entity Recognition (NER), document classification, semantic search, knowledge graph–based relationship mapping, rule-based reasoning, information extraction, recommendation engines, predictive analytics, workflow intelligence, explainable AI with source citations, and dashboard analytics.
Measured outcome (accuracy, rework reduction, time saved, etc.)	Automated manual document interpretation, compliance verification, and engineering decision-support workflows, significantly reducing manual effort, improving consistency of rule validation, accelerating decision-making, increasing traceability, and enabling AI-assisted impact assessment across interconnected business processes.
Demonstrable? (Live demo ☐ Video ☐ Reference call ☐)	Reference call
Evidence attached	Synergy Medtech

Field	Response
Client / End-User	The GoPT Physiotherapy
Problem solved (map to this challenge)	Designed and deployed a centralized operations management platform to digitize workflow execution, resource planning, scheduling, and process coordination across multiple branches. The solution managed interdependent operational activities, maintained data

	consistency, automated approval workflows, and improved organizational visibility. Similar to the MDL challenge, it addressed dependency management, process traceability, workflow optimization, and decision support within a complex multi-user environment.
Input data types, formats and sources handled as a part of similar projects.	Structured operational data including appointment schedules, resource availability, customer records, workflow data, branch information, business documents, spreadsheets (XLS/XLSX/CSV), PDFs, forms, relational databases, and user-generated records integrated through web applications and backend databases.
AI/CV techniques applied	Workflow automation, business rule engines, scheduling optimization algorithms, decision-support logic, operational analytics, reporting dashboards, role-based workflow management, and process monitoring.
Measured outcome (accuracy, rework reduction, time saved, etc.)	Replaced manual operational workflows with a centralized digital platform, improving scheduling efficiency, reducing administrative overhead, increasing process visibility, enhancing resource utilization, and enabling faster, data-driven operational decision-making across multiple locations.
Demonstrable? (Live demo ☐ Video ☐ Reference call ☐)	Reference call
Evidence attached	GOPT

Field	**Response**
Client / End-User	JR Engineering Works
Problem solved (map to this challenge)	Developed and deployed a digital Workshop Management System to streamline engineering workshop operations, production planning, inventory management, job tracking, material traceability, and workflow execution. Similar to the MDL challenge, the solution managed interconnected engineering processes, synchronized production activities with material availability, improved operational visibility, and supported management decision-making through centralized monitoring and reporting.
Input data types, formats and sources handled as a part of similar projects.	Structured engineering and operational data including production work orders, inventory records, material consumption logs, customer orders, job cards, invoices, operational logs, spreadsheets (CSV/XLS/XLSX), PDFs, user-entered forms, and relational database transactions integrated through the workshop management platform.

AI/CV techniques applied	Workflow automation, business rule engines, production workflow optimization, inventory tracking algorithms, operational analytics, KPI monitoring, reporting dashboards, and decision-support mechanisms.
Measured outcome (accuracy, rework reduction, time saved, etc.)	Digitized manual workshop operations through a centralized platform, improving inventory accuracy, enhancing material traceability, enabling real-time production monitoring, reducing administrative effort, streamlining production and billing workflows, and providing management with actionable operational insights for faster decision-making.
Demonstrable? (Live demo ☐ Video ☐ Reference call ☐)	Reference call
Evidence attached	JR Engineering

Field	**Response**
Client / End-User	Mr. Ranjeet Rajan, Patent Lawyer
Problem solved (map to this challenge)	Developed an AI-powered prior-art intelligence platform that automated analysis of large volumes of technical documents to identify relevant prior art, extract engineering knowledge, discover relationships between technical concepts, assess the impact of new inventions on existing technologies, and generate explainable recommendations. Similar to the MDL challenge, the solution performed intelligent document understanding, dependency mapping, semantic impact analysis, traceable decision support, and knowledge-driven engineering analysis across interconnected technical artefacts.
Input data types, formats and sources handled as a part of similar projects.	Patent specifications, technical documents, engineering publications, research papers, invention disclosures, patent claims, citation records, CPC/IPC classification data, PDF documents, XML patent records, structured metadata, and technical text sourced from Google Patents, Espacenet, WIPO PATENTSCOPE, USPTO, InPASS, and The Lens.
AI/CV techniques applied	Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Named Entity Recognition (NER), semantic search, document classification, information extraction, vector embeddings, knowledge graph construction, citation analysis, similarity ranking, recommendation engines, explainable AI with source traceability, and dashboard analytics.
Measured outcome (accuracy, rework	Reduced prior-art discovery time from hours of manual research to minutes through AI-assisted semantic retrieval, improved search relevance using embedding-based similarity matching, automated ranking and filtering of thousands of technical documents, enhanced

reduction, time saved, etc.)	traceability through citation-based evidence, and significantly reduced manual research and document review effort.
Demonstrable? (Live demo ☐ Video ☐ Reference call ☐)	Video
Evidence attached	Prior-Art Video

5. Understanding of the Problem Statement (Descriptive)

Provide clear, specific answers (this section + Section 6 together ≤ 10 pages).

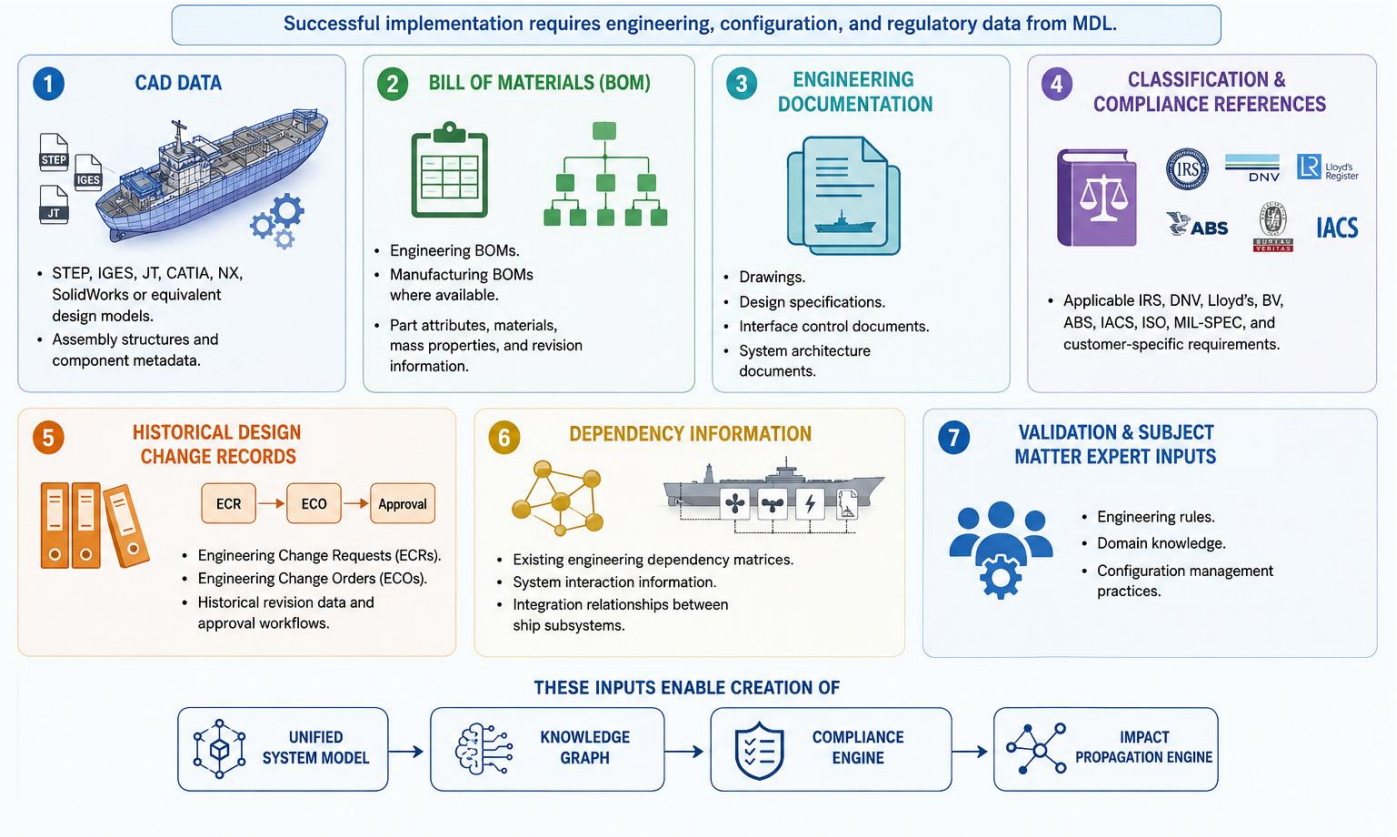
1. **In your own words, describe the underlying problem and the real operational pain it causes in a shipyard.**

- **Engineering changes rarely remain isolated.** A modification to one component (equipment, pipe routing, structural member, cable route, etc.) often affects multiple engineering disciplines, making it difficult to identify every downstream dependency manually.
- **Impact assessment depends heavily on engineering experience.** Engineers must review numerous CAD models, drawings, BOMs, standards, and historical documents to determine what else needs modification, making the process slow and susceptible to missed dependencies.
- **Maintaining consistency between 3D models and the Bill of Material (BOM) is a continuous challenge.** Manual updates frequently result in missing components, incorrect quantities, specification mismatches, or outdated BOM entries, creating issues during procurement, fabrication, and assembly.
- **Compliance verification is time-consuming.** Engineers must manually verify that every design revision satisfies applicable classification society rules, standards, and regulations before submission, increasing engineering effort and review cycles.
- **Late identification of design conflicts increases rework.** If impacts or compliance issues are discovered after drawings are released or manufacturing has begun, engineering teams must revise designs, update documentation, repeat approvals, and sometimes modify completed work.
- **Configuration control becomes increasingly difficult as projects progress.** Multiple drawing revisions, design changes, and approvals must remain synchronized across disciplines to ensure production always uses the latest approved configuration.
- **The cumulative operational impact is increased engineering effort, longer design cycles, avoidable rework, schedule delays, higher project costs, and greater risk of inconsistencies reaching production.**

2. **Who are the stakeholders and how will each interact with your solution?**

Stakeholder	Role in MDL	How They Interact with AURORA
Detail Design Engineers (Hull, Structural, Propulsion, Electrical, Weapons & Auxiliary Systems)	Develop and modify detailed ship designs while ensuring compliance with class rules and engineering standards.	Upload and update CAD models, review AI-generated compliance checks, receive optimization recommendations, identify 3D–BOM mismatches, analyze downstream design impacts, and implement corrective actions before design submission.
Design Managers / Chief Designer	Review and approve engineering decisions, resolve cross-disciplinary conflicts, and finalize ship designs before release.	Monitor design status through dashboards, review AI-generated impact assessments, evaluate optimization recommendations, approve or reject engineering changes, and authorize design baselines for production.
Classification & Regulatory Liaison	Ensure designs comply with classification society rules, naval standards, and statutory regulations before submission for approval.	Validate AI-generated compliance reports with clause-level references, verify regulatory compliance, prepare submission packages, review non-compliance findings, and monitor approval status throughout the design lifecycle.
Configuration & Revision Control Engineers	Maintain version-controlled engineering data, design revisions, and drawing releases throughout the project lifecycle.	Manage revision history, compare design versions, synchronize approved changes between CAD models and BOM, maintain complete traceability, and control release of engineering baselines.
Production Planning & Drawing Office	Convert approved engineering designs into production-ready drawings and manufacturing documentation.	Verify that released drawings correspond to approved revisions, ensure BOM consistency before manufacturing, identify unresolved discrepancies, and generate production-ready documentation.
Quality Assurance (QA) Engineers	Verify engineering quality, compliance, and traceability before design release.	Review compliance reports, validate engineering evidence, audit design decisions, verify BOM reconciliation results, and generate quality assurance and verification reports.
IT & Cybersecurity Administrators	Deploy, secure, and maintain the platform within MDL's air-gapped infrastructure	Configure user roles and permissions, integrate with PLM/CAD/ERP systems, monitor audit logs, manage system security, perform backups, and ensure secure on-premise operation.

3. **What input data, document or information required from MDL for the successful execution?**



4. **How does the proposed system address variations, exceptions and deviations identified during the execution**

   AURORA is designed to operate in dynamic engineering environments where variations, exceptions, and deviations are expected throughout the ship design lifecycle.

   The platform maintains a dependency-aware Unified System Model and Knowledge Graph that continuously captures relationships among components, systems, documents, and compliance requirements.

   The Compliance Engine performs rule-based validation against applicable standards and identifies any resulting compliance deviations. Variations are classified based on severity, dependency impact, compliance implications, and engineering risk.

   Where exceptions require engineering judgment, the system supports controlled workflows for review, approval, justification, and traceability. All deviations are recorded in immutable audit logs, ensuring complete visibility of decision-making throughout the design lifecycle.

   This approach allows MDL to manage engineering changes systematically while maintaining configuration integrity and regulatory compliance.

5. **How does the proposed solution handle the incomplete, inconsistent or complex input conditions?**

   AURORA combines deterministic validation, confidence scoring, and engineering review workflows to handle incomplete, inconsistent, or complex data conditions.

The system performs automated data validation during ingestion to identify missing attributes, conflicting information, duplicate records, unsupported formats, and inconsistent engineering references.

For CAD-BOM synchronization, multiple matching signals including geometry similarity, naming similarity, mass comparison, and assembly position are evaluated. An ML-based confidence scoring mechanism assesses match quality and identifies uncertain cases requiring engineering review.

Where information is incomplete, the platform continues operating using available validated data while clearly flagging missing dependencies, unresolved references, and confidence limitations. Complex engineering relationships are represented through the Knowledge Graph, allowing the system to reason across interconnected data sources.

The solution never makes unverified engineering decisions automatically. Low-confidence scenarios are escalated to authorized users for validation, ensuring reliability and engineering accountability.

6. **How does the system manage revisions, versions and baseline information for accurate comparison and analysis?**
AURORA incorporates robust configuration management principles within its Unified System Model architecture to maintain complete revision, version, baseline, and engineering traceability throughout the ship design lifecycle.

Each engineering object, including components, assemblies, CAD models, documents, BOM entries, drawings, simulation references, and compliance records is uniquely identified and maintains a complete version history with associated metadata such as author, timestamp, revision reason, approval status, and engineering discipline.

The platform utilizes a local Git-based version control system (deployed entirely within MDL's secure air-gapped environment) to track engineering artifacts, ensuring every design change is versioned, traceable, and recoverable. Approved milestones are captured as immutable design baselines, while labels/tags are used to identify significant project stages such as Contract Design, Basic Design, Class Submission, Class Approved, and Production Release. Older revisions and completed baselines are securely archived while remaining fully accessible for audit, comparison, and regulatory compliance.

Whenever a revision occurs, AURORA automatically detects differences between versions, identifies affected components and dependencies, evaluates downstream engineering impacts, and generates comprehensive impact assessment reports.

Version comparison capabilities enable engineers to analyze changes at the component, assembly, subsystem, and complete ship levels, with side-by-side visualization of geometry, BOM modifications, document revisions, compliance status, and dependency changes. Historical baselines remain preserved to support audits, compliance verification, rollback, and complete engineering traceability.

This ensures that all impact analysis, compliance verification, optimization activities, and approval workflows are always performed against the correct approved design configuration while maintaining a complete and auditable engineering history.

7. **How does the system identify, analyse and report variations or deviations from the defined requirements?**

   AURORA continuously compares engineering data against approved requirements, baselines, and compliance rules to identify deviations.

   The Compliance Engine evaluates designs against classification rules, standards, specifications, and project-specific requirements. Simultaneously, the CAD-BOM Synchronization Engine detects inconsistencies between design models and associated material definitions.

   When a deviation is detected, the Impact Propagation Engine analyzes affected systems, determines the scope of influence, and calculates dependency-based impact scores. The system then categorizes findings according to severity, affected disciplines, compliance implications, and engineering risk.

   Reports include:

   - Description of the deviation.

   - Affected components and systems.

   - Root dependency path.

   - Compliance impact.

   - Recommended corrective actions.

   - Associated revision information.

   Results are presented through dashboards, audit reports, and engineering review workflows, enabling rapid identification and resolution of issues.

8. **State your key assumptions, clarification questions, and dependencies on the shipyard.**

   Key Assumptions
   1. Relevant CAD models, BOMs, engineering documents, and revision histories will be available for integration.
   2. Applicable classification rules and standards can be provided or referenced by MDL.
   3. Subject Matter Experts (SMEs) will be available for validation of engineering dependencies and business rules.
   4. Necessary IT infrastructure will be available for deployment within MDL's secure environment.
   5. Required approvals will be available for access to engineering and configuration data.

   Clarification Questions
   1. Which CAD platforms and versions are currently used across MDL projects?
   2. What level of BOM maturity and standardization exists across programs?
   3. Which classification societies and defence standards are applicable for the target projects?
   4. What existing configuration management and engineering change processes are currently followed?

5. What level of integration is expected with existing PLM, ERP, CAD, and document management systems?

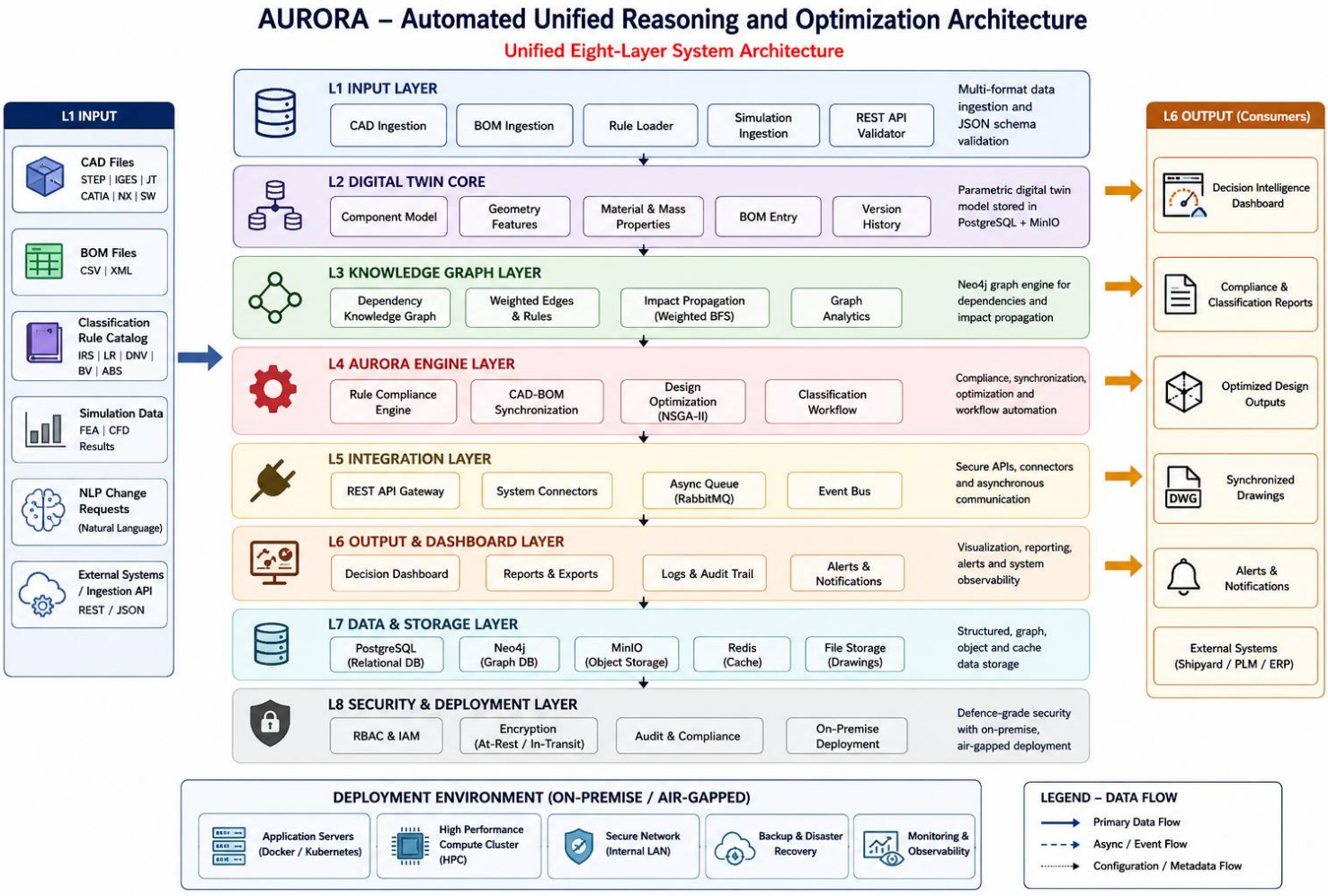
Dependencies on the Shipyard

1. Availability of engineering data and historical records.
2. Availability of domain experts for validation and acceptance testing.
3. Access to relevant standards and compliance requirements.
4. Support for integration with existing engineering systems.
5. Participation in pilot validation activities and operational evaluation.

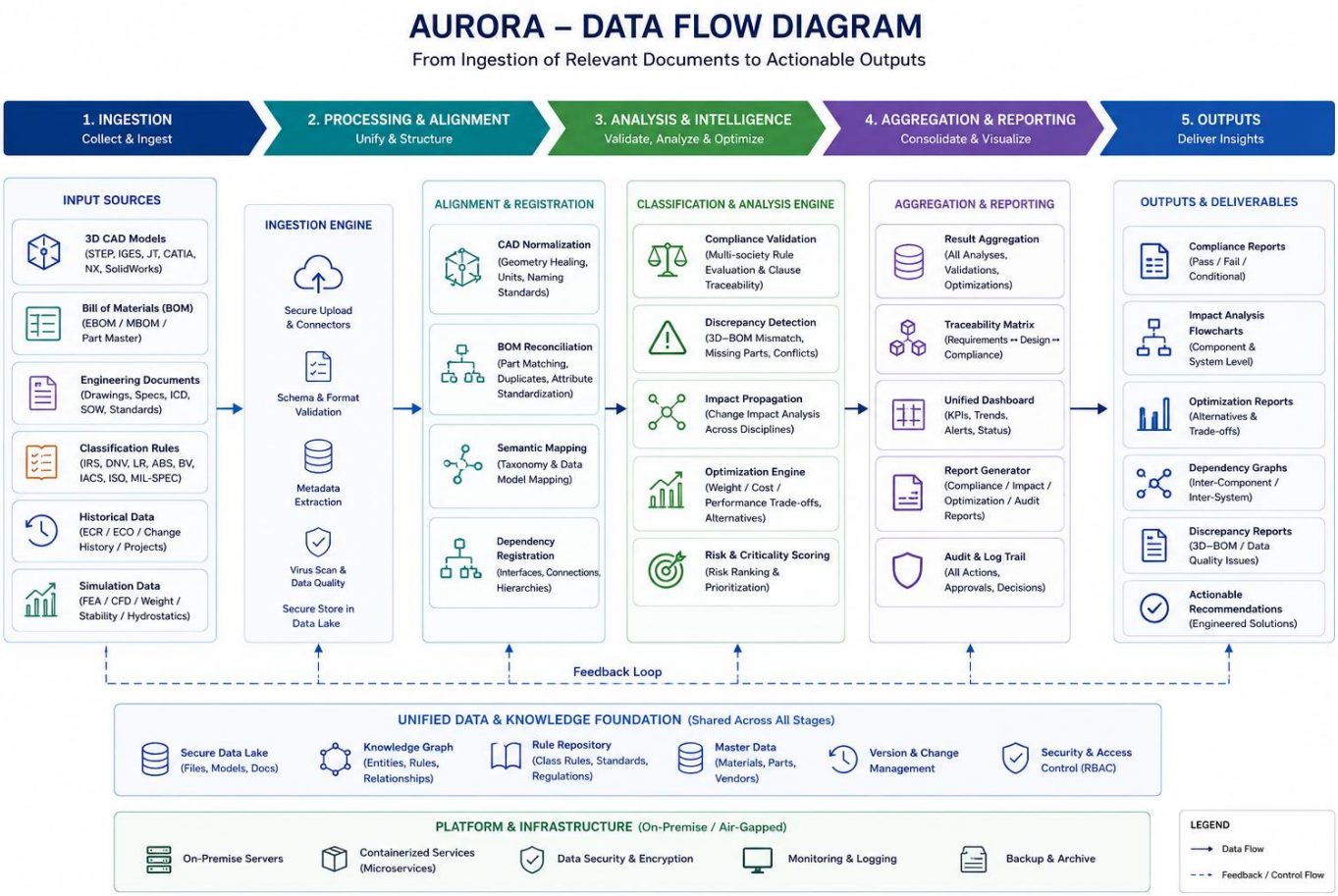
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6. Solution Architecture & Data Flow

1. **Provide a solution architecture diagram (Annexure) covering ingestion → alignment/registration → classification → reporting → design-department hand-off.**



2. **Provide a data-flow diagram from input (ingestion of relevant documents) to outputs (reports/flowcharts/graphs).**



3. **Describe how you ingest and interpret 2D production drawings and 3D CAD models (formats supported , e.g., PDF, TIFF, DWG, DXF, STEP, IFC, IGES, native 3D).**

AURORA provides a unified engineering data ingestion pipeline for both 2D engineering documentation and 3D CAD models.

For 2D INPUT

Supported 2D inputs include PDF, TIFF, DWG and DXF drawings. Where required, OCR is applied to scanned drawings to extract engineering text, annotations and metadata. Structured information such as drawing numbers, revisions, dimensions and references is linked to the corresponding engineering objects within the Unified System Model.

FOR 3D INPUT

Supported 3D formats include STEP, IGES, JT and native CAD models from CATIA, Siemens NX and SolidWorks through standard exchange mechanisms. The CAD Parsing Engine, based on OpenCascade, extracts assembly hierarchies, geometry, topology, materials, mass properties and component metadata.

All ingested information is normalized into a common Unified System Model model, where components are linked with BOM items, engineering documents, revision history and compliance requirements. This enables downstream CAD–BOM synchronization, impact analysis and rule-based compliance validation.

4. **Describe how you ingest and process data (point cloud formats e.g. E57/LAS/PTS, photogrammetry, images) and how you register it to the design coordinate system.**
   The core scope of AURORA is centered on the ingestion and processing of engineering design data, including 3D CAD models, Bills of Material (BOM), engineering drawings, specifications, standards, and associated design documentation. These inputs are normalized, validated, and integrated into the Unified System Model to enable design optimization, compliance analysis, and 3D–BOM synchronization.

   The current proposed solution does not rely on point cloud processing or photogrammetry, as these capabilities are outside the scope of the present problem statement and are not required for achieving the objectives of design optimization and CAD–BOM synchronization.

   However, the solution has been designed with a modular architecture that can be extended, if required by MDL in the future, to support as-built verification workflows. Such an extension can ingest industry-standard point cloud formats (e.g., E57, LAS, PTS), photogrammetry outputs, and engineering images. These datasets can be pre-processed, aligned with the ship design coordinate system using survey reference points or common geometric features, and compared with the digital design model to identify deviations.

   This capability would be offered as an optional enhancement and can be integrated without requiring significant changes to the core AURORA platform, should future operational requirements include digital inspection or as-built validation.

5. **Describe the solution finding approach (positional/relocation deviation, dimensional deviation, tolerance thresholds).**
   AURORA evaluates engineering deviations using deterministic rule-based analysis combined with dependency-aware impact assessment.
   The solution supports:
   - **Positional deviations** by comparing component placement against approved design coordinates.
   - **Dimensional deviations** by comparing extracted geometric properties with approved design specifications.
   - **CAD–BOM deviations** by identifying missing, extra, mismatched, or inconsistent components.
   - **Compliance deviations** by validating design parameters against applicable class rules, standards, and regulations.

   Tolerance limits are fully configurable and can be defined globally, per engineering discipline, or per component type based on MDL engineering practices and applicable classification society requirements. The specific tolerance thresholds, validation criteria, deviation severity levels, and acceptance limits will be finalized through mutual agreement with MDL during the implementation phase to align with the shipyard's engineering standards, project requirements, and operational workflows.
   Detected deviations are analyzed by the Knowledge Graph and Impact Propagation Engine to determine affected assemblies, downstream engineering impacts, and compliance implications before recommending engineering review and corrective actions.

6. **Describe the reporting output (visual overlay, deviation list, severity, coordinates) and the hand-off mechanism to the design department.**
   The system generates comprehensive engineering reports through role-based dashboards and exportable documents.

   Reporting includes:
   • Visual comparison of affected CAD assemblies and engineering objects.
   • List of detected deviations and affected components.
   • Severity classification (Critical, Major, Minor or Informational).
   • Associated engineering coordinates, assemblies and subsystem references where applicable.
   • Rule or standard references for compliance-related findings.
   • CAD–BOM mismatch summaries.
   • Downstream dependency and impact analysis.
   • Recommended engineering actions.

   Reports can be exported in PDF and standard interchange formats and are integrated into the engineering review and approval workflow. Approved changes are linked to revision-controlled Unified System Model objects and handed over to the design department for incorporation into basic drawings prior to class submission and subsequently into production drawings after approval, maintaining complete auditability and configuration traceability.

7. **Describe integration with existing CAD / PLM / design systems (name systems and interfaces/APIs).**
   AURORA is designed as an open, standards-based platform that integrates with existing engineering ecosystems rather than replacing them.
   The platform supports integration with major CAD systems including CATIA, Siemens NX, SolidWorks, AutoCAD and other applications through standard CAD exchange formats such as STEP, IGES and JT.

   Integration with PLM and engineering information systems such as Teamcenter, ENOVIA and Windchill is supported through REST APIs, standard data exchange interfaces and configurable connectors. BOM information can also be imported from ERP, PLM and spreadsheet-based systems.

   The architecture exposes secure REST APIs for interoperability with shipyard applications while preserving revision history, configuration management, audit trails and role-based access control. This enables seamless synchronization between CAD models, BOMs, engineering documents and downstream workflows without vendor lock-in.

8. **Describe scalability , how the architecture performs across a full ship and across projects (thousands to millions of drawings/scan tiles).**
   AURORA is designed as a modular, service-oriented architecture capable of scaling from individual subsystems to complete ship platforms and multiple concurrent projects.

Engineering data is organized within a Unified System Model and Neo4j-based Knowledge Graph, allowing dependency analysis to operate only on affected portions of the design instead of the complete model. The Impact Propagation Engine performs graph-based traversal using weighted relationships and depth-aware propagation, enabling efficient analysis even for highly interconnected systems.

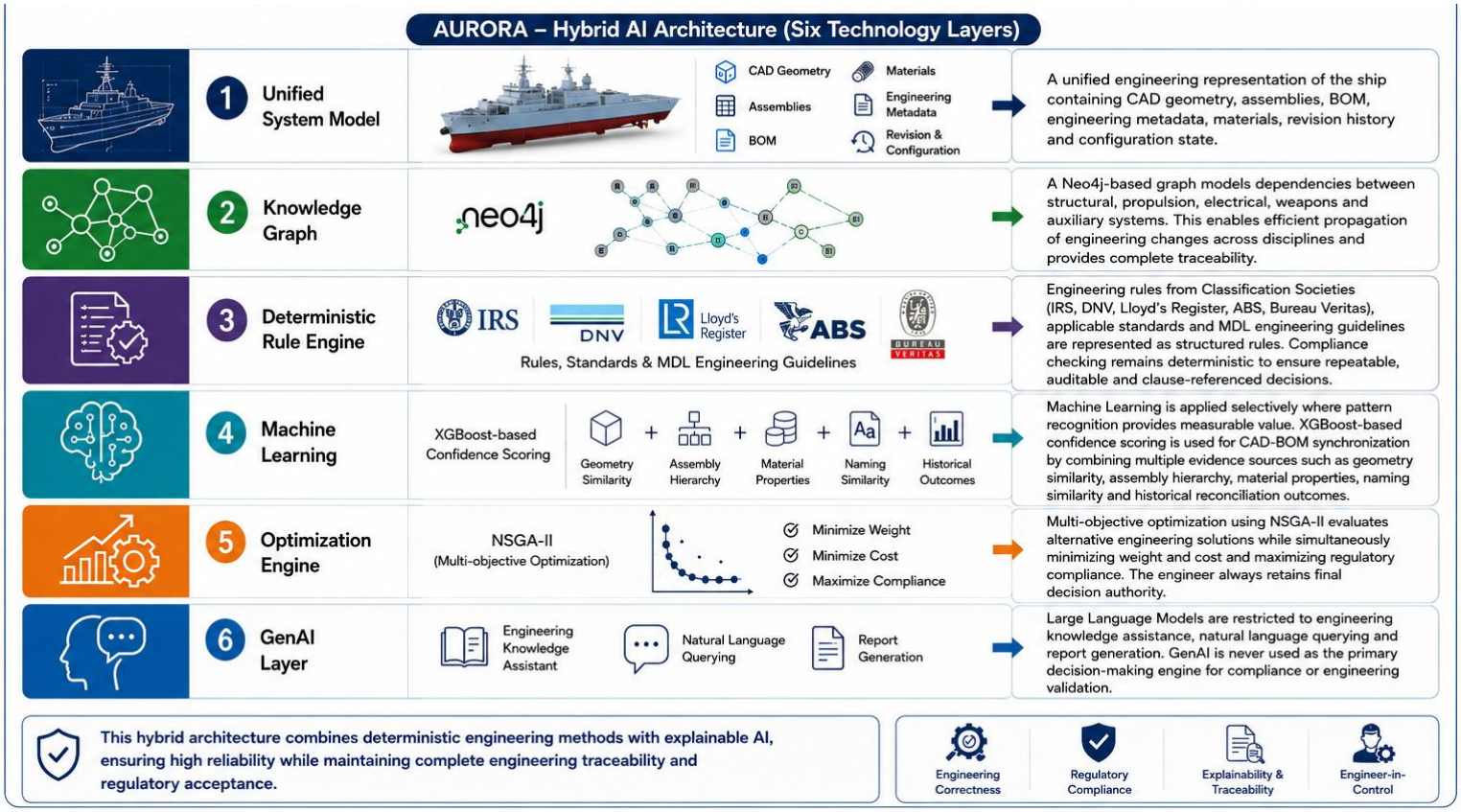
Independent services for CAD parsing, BOM synchronization, compliance validation, optimization and reporting can execute in parallel, supporting batch processing of large engineering datasets.

The architecture is designed to scale across thousands of drawings, large BOMs, extensive CAD assemblies and multi-disciplinary engineering data while maintaining revision traceability, auditability and performance without requiring architectural redesign. This aligns with MDL's requirement to support large rule, BOM and model volumes across multiple shipbuilding projects.

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7. AI / GenAI Technical Capability (Descriptive)

1. **Which technologies or innovations approach proposed for addressing the problem and justify the selection of models, techniques and methodologies.**



2. **Where (if at all) do LLMs / GenAI / RAG add value in this solution (e.g., report narration, rule interpretation, query interface)? Be specific , do not over-claim GenAI where classical CV/3D is more appropriate.**

LLM / GenAI / RAG Capability	Why This Was Used	Benefits
Natural Language Query Interface	Enables engineers to interact with the system using plain English instead of navigating multiple engineering tools and reports.	Improves usability, reduces search time, and allows engineers to quickly obtain design insights such as affected systems, non-compliance reasons, and impacted compartments.
Rule Interpretation	Classification Society rules and engineering standards are often lengthy and difficult to interpret. LLMs convert them into engineer-friendly explanations while always displaying the original cited clause.	Faster understanding of complex regulations, reduced interpretation effort, improved productivity, while maintaining traceability and regulatory compliance.
Retrieval-Augmented Generation (RAG) for Engineering Knowledge Search	Retrieves relevant information from Classification Society rules, standards, engineering guidelines, historical revisions, and design documentation before generating responses.	Ensures responses are grounded in authoritative documents, minimizes hallucinations, improves accuracy, and provides citation-backed answers.
Automated Report Generation	Generates engineering documentation from deterministic analysis outputs rather than manually compiling reports.	Produces executive summaries, compliance reports, impact assessment reports, and engineering recommendations quickly while maintaining consistency and reducing documentation effort.
Interactive Engineering Assistant	Provides contextual explanations for dependency propagation, revision history, engineering decisions, and BOM inconsistencies using information generated by deterministic modules.	Improves decision support, simplifies knowledge retrieval, accelerates troubleshooting, and enhances engineer productivity without replacing engineering judgment.

3. **Explain your RAG architecture (if used): retrieval source, chunking, embedding model, grounding, and how hallucinations are minimised.**
   The Retrieval-Augmented Generation (RAG) architecture is designed as an engineering knowledge retrieval system rather than a decision-making engine.

   Knowledge Sources
   - Classification Society rule books
   - National and international engineering standards

• MDL engineering procedures
• Historical approved engineering revisions
• Design documentation
• Technical manuals
• BOM metadata
• Unified System Model metadata

Document Processing
Engineering documents are parsed, OCR processed where required and segmented into semantically meaningful sections such as clauses, chapters, tables and engineering specifications. Each chunk preserves document identifiers, section numbers, revision information and source references.

Embedding Model
Sentence Transformers (all-mpnet-base-v2) generate semantic embeddings for all engineering documents.

Vector Database
Embeddings are indexed in Qdrant for semantic retrieval with associated metadata including document version, discipline, revision and source.

Retrieval Process
When an engineer submits a query:
1. Query embedding is generated.
2. Relevant engineering documents are retrieved from the vector database.
3. Retrieved context is validated using metadata filters.
4. Only retrieved context is passed to the LLM.
5. Generated responses always include references to the originating engineering documents.

Hallucination Mitigation
• Retrieval is restricted to approved engineering documents.
• Every generated answer cites its source documents.
• Low-confidence retrieval returns "Insufficient evidence" instead of speculative responses.
• Rule citations remain visible to the user.
• Engineering decisions continue to be produced by deterministic rule engines rather than the LLM.

This architecture ensures that GenAI functions purely as an intelligent engineering assistant while all engineering validation remains evidence-based and auditable.

4. **Explain your confidence scoring mechanism for each detected deviation.**
Every detected engineering observation is assigned a confidence score representing the quality and consistency of supporting evidence.

For CAD-BOM synchronization, confidence is computed using a weighted combination of:
• Geometry similarity
• Assembly hierarchy consistency
• Component naming similarity
• Material and specification matching
• Mass and dimensional consistency

• Spatial position consistency
• Historical matching behaviour

The weighted features are evaluated using an XGBoost classifier trained on validated engineering reconciliation examples.

For compliance validation, confidence is derived from deterministic factors including:
• Rule matching completeness
• Parameter availability
• Input data quality
• Rule ambiguity
• Number of supporting engineering constraints satisfied

Confidence is presented as High, Medium or Low together with the underlying evidence.

Low-confidence results are automatically routed for engineering review before approval.

5. **Explain how false positives are reduced and how detection accuracy is measured (define your metrics , precision/recall/IoU/etc.).**
   False positives are minimized through a combination of deterministic engineering validation, multiple independent evidence sources and configurable engineering tolerances.

   Reduction mechanisms include:
   • Cross-validation using multiple engineering attributes.
   • Configurable discipline-specific tolerance thresholds.
   • Rule consistency verification.
   • Revision-aware comparison.
   • Duplicate detection suppression.
   • Human validation before final approval.

   Model performance is continuously evaluated using labelled engineering validation datasets. Metrics include:
   • Precision – percentage of flagged deviations that are genuinely incorrect.
   • Recall – percentage of actual deviations successfully detected.
   • F1 Score – balanced measure combining precision and recall.
   • CAD-BOM Match Accuracy.
   • Compliance Detection Accuracy.
   • False Positive Rate.
   • False Negative Rate.
   • Mean Processing Time.

   For geometry-based comparison, dimensional error distributions and coordinate deviations are also measured where applicable.

   Performance targets are jointly established with MDL during pilot deployment and continuously monitored through validation dashboards.

6. **Explain explainability / traceability , how a user sees *why* a deviation was flagged.**
   Explainability is a fundamental design principle of AURORA.

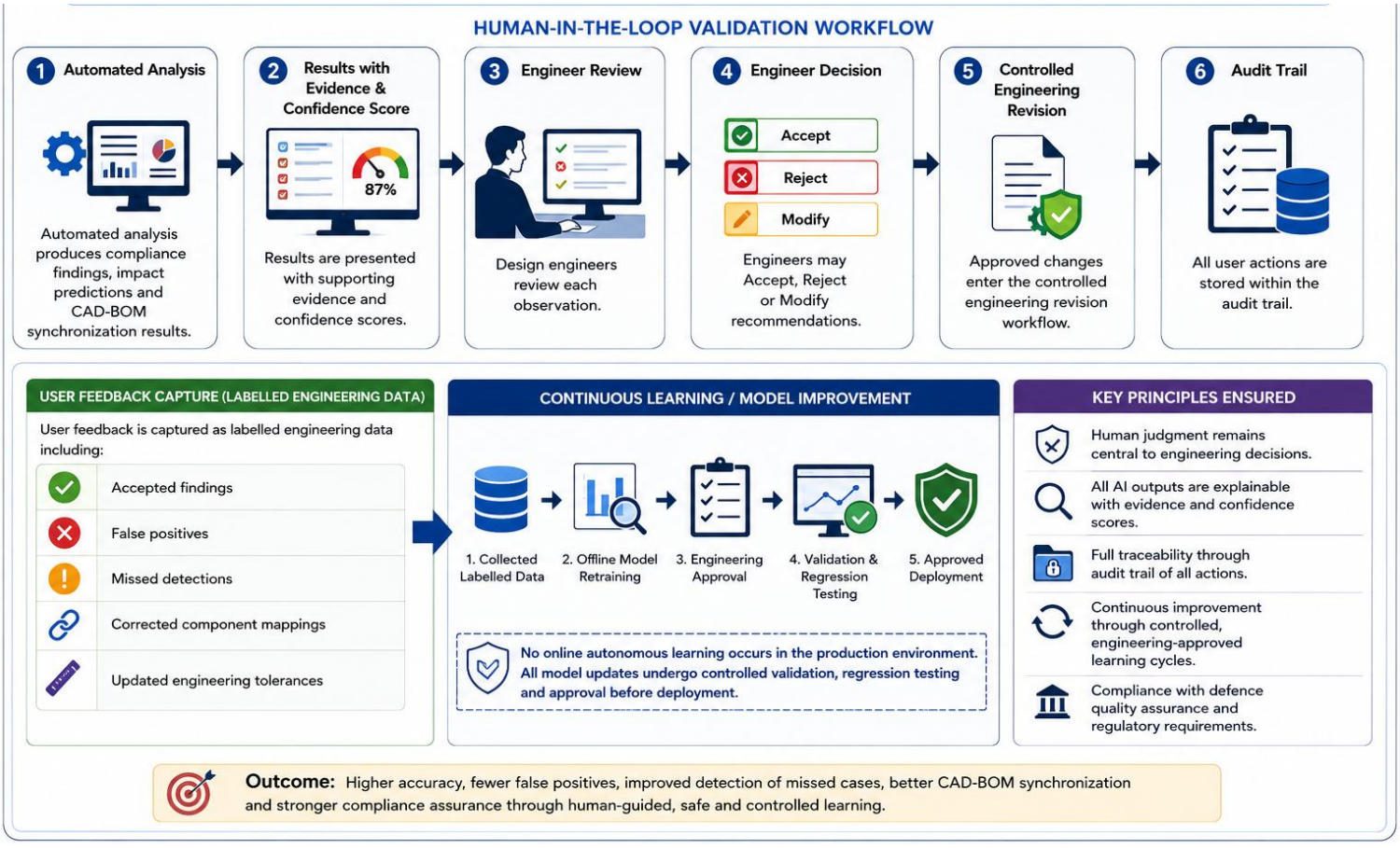
   Every engineering finding is accompanied by complete traceability showing:

- The affected engineering object.
- Associated CAD component.
- Linked BOM item.
- Applicable engineering rule or standard.
- Specific clause reference.
- Input parameters used.
- Intermediate validation steps.
- Final engineering conclusion.
- Confidence score.
- Version and revision identifiers.

The Knowledge Graph maintains end-to-end traceability between CAD models, engineering rules, BOM entries, design revisions and impacted downstream systems. Users can drill down from a high-level dashboard directly to the supporting engineering evidence, enabling transparent engineering review and audit.

No compliance decision is presented without supporting engineering evidence and source references.

7. **Explain the human-in-the-loop validation workflow and how user corrections feed continuous learning / model improvement.**



8. **Explain your model training / fine-tuning strategy and training-data requirements, including how you cope with limited/sensitive shipyard data (synthetic data, transfer learning, few-shot).**

AURORA adopts a controlled, offline model development strategy aligned with defence quality assurance practices. The platform remains fully functional using deterministic engineering rules and knowledge graph reasoning, while Machine Learning components are trained only where measurable performance improvements are demonstrated.

Training Data Requirements

The ML models are trained using validated engineering datasets, including:

- Historical CAD–BOM reconciliation records
- Engineering Change Requests (ECR/ECO)
- Design revision history
- Approved BOM modifications
- Compliance review outcomes
- Historical design change propagation records
- Engineer-validated feedback from production deployments

Model Training Strategy

- **Data Split:** Engineering datasets are divided into **70% Training, 15% Validation, and 15% Test** sets while ensuring revision histories from the same project are not split across datasets to prevent data leakage.
- **Cross Validation: 5-Fold Cross Validation** is used during model development to evaluate model stability and improve generalization, particularly when labelled datasets are limited.
- **Hyperparameter Optimization:** Model parameters are optimized using Grid Search or Bayesian Optimization based on validation performance.
- **Performance Evaluation:** Models are evaluated using task-specific metrics such as Precision, Recall, F1-Score, ROC-AUC (where applicable), False Positive Rate, and engineering acceptance criteria agreed with MDL.
- **Model Selection:** Only models demonstrating consistent performance across validation folds and meeting predefined engineering accuracy thresholds are considered for deployment.

Limited Data Strategy

To address limited labelled shipyard data during initial deployment, the following techniques are employed:

- Transfer Learning using pre-trained engineering language models
- Synthetic engineering datasets generated through controlled rule and design variations
- Engineering data augmentation using valid geometric and metadata transformations
- Few-shot learning for new equipment categories and engineering domains
- Active Learning using engineer-reviewed corrections to prioritize future training samples

Model Validation & Deployment

- Training and fine-tuning are performed entirely within MDL's secure on-premise infrastructure.
- No production data leaves the shipyard environment.

- All trained models undergo independent validation, regression testing, bias assessment, and engineering review before deployment.
- Models are version-controlled with complete traceability of training data, parameters, evaluation metrics, and approval records.
- **No online or autonomous learning occurs in production.** Model updates are performed only through controlled offline retraining and formal engineering approval.

This strategy ensures that AI models remain accurate, explainable, reproducible, and compliant with defence quality assurance and configuration management practices.

9. **How are engineering rules / tolerances / classification-society requirements incorporated?**

Engineering rules are represented as structured, version-controlled knowledge assets within the Compliance Engine.

Supported rule sources include:
• Indian Register of Shipping (IRS)
• DNV
• Lloyd's Register
• American Bureau of Shipping (ABS)
• IACS Unified Requirements
• ISO Standards
• MIL-STD requirements
• MDL engineering procedures

Rules are transformed into machine-readable representations while preserving original clause references, revision identifiers and applicability conditions.

Engineering tolerances are configurable at multiple levels:
• Project level
• Discipline level
• Equipment type
• Component category
• Individual engineering parameter

Every compliance evaluation references the exact rule, standard and clause responsible for the determination.

The rule repository supports version management, allowing updated Classification Society revisions to be incorporated without changes to application source code. Historical rule versions remain available to preserve complete auditability of earlier engineering decisions.

This deterministic rule-based approach ensures that compliance assessments remain transparent, repeatable and fully traceable while allowing controlled updates as regulations evolve.

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8. Technology Stack & AI Model Disclosure

8.1 Technology Stack

Layer	Technology / Product	Version	Open Source / Commercial	Country of Origin	License Type
Programming Language(s)	Python TypeScript	3.12 LTS 5.8	OpenSource OpenSource	Netherlands USA	PSF License Apache 2.0
Frontend	React	1.9.x	OpenSource	USA	MIT
Backend	FastAPI Uvicorn SQLAlchemy	0.116+ 0.35 2.0	OpenSource OpenSource OpenSource	USA USA USA	MIT BSD MIT
AI/ML Framework	Pytorch Scikit-learn	2.7 1.7	OpenSource OpenSource	USA France	BSD BSD
Computer Vision Framework	OpenCV Segment Anything 2 (SAM2)	4.12 LTS	OpenSource Opensource	USA USA	Apache 2.0 Apache 2.0
3D / Point Cloud Library	Open3D PDAL VTK	0.19 2.9 9.x	OpenSource OpenSource OpenSource	USA USA USA	MIT BSD BSD
OCR Engine	Tesseract OCR	5.5	OpenSource	USA	Apache 2.0
LLM / GenAI Model	Llama 3.1 8B Instruct (Offline)	Latest	OpenWeights	USA	Meta Community License
Embedding Model	all-mpnet-base-v2	v.2	OpenSource	UK	Apache 2.0
Vector Database	Qdrant	1.5	OpenSource	Poland	Apache 2.0
Knowledge Graph (if any)	ArcadeDB	25.x	OpenSource	Italy	Apache 2.0
Primary Database	PostgreSQL	17	OpenSource	USA	PostgreSQL Licence
Cache / Queue	Redis	8	OpenSource	USA	RSALv2 / SSPLv1

Containerisation / Orchestration	Docker Engine Kubernetes	28 1.33	OpenSource OpenSource	USA USA	Apache 2.0 Apache 2.0
Operating System	Ubuntu Server LTS	24.04	OpenSource	UK	GPL
Third-Party / External Dependencies	spaCy Hugging Face ONNX Runtime Alembic Pydantic Sentence Transformers Drools Rule Engine	3.8 4.53 1.22 1.16 2.11 5.x 10.x	Open Source OpenSource OpenSource Opensource OpenSource OpenSource OpenSource	Germany France USA USA USA UK USA	MIT Apache 2.0 MIT MIT MIT Apache 2.0 Apache 2.0

8.2 AI Model Disclosure (Mandatory , one row per model/component)

Category	**Product / Model Used**	**Version**	**Commercial / Open Source**	**Country of Origin**	**Hosted Where (Local / Cloud)**	**License**
LLM / GenAI	Llama 3.1 8B Instruct	LTS	Open Weights	USA	Local	Meta Llama 3.1 Community License
OCR	Tesseract OCR	5.5	OpenSource	USA	Local	Apache 2.0
Embedding	all-mpnet-base-v2 (Sentence Transformers)	v.2	OpenSource	UK	Local	Apache 2.0
Object Detection	YOLOv11	LTS	OpenSource	USA	Local	AGPL-v3
Segmentation	Segment Anything Model 2 (SAM 2)	LTS	OpenSource	USA	Local	Apache 2.0
3D / Point Cloud	Open3D	0/19	OpenSource	USA	Local	MIT
Vector DB	Qdrant	1.15	OpenSource	Poland	Local	Apache 2.0

Knowledge Graph	ArcadeDB	25.x	OpenSource	Italy	Local	Apache 2.0
Image Processing	OpenCV	4.12	OpenSource	USA	Local	Apache 2.0
Any other AI component	Drools Rule Engine (Deterministic Rule Engine)	10.x	OpenSource	USA	Local	Apache 2.0

Why this is asked: A structured disclosure of *exactly* which models/products are used (with version, origin and hosting) is required for security clearance and for verifying air-gapped feasibility. Generic answers such as "an LLM" or "a vision model" will be scored as "not disclosed".

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9. Security, Data Sovereignty & Air-Gapped Readiness

#	Question	Response
9.1	Can the solution be deployed fully **on premises**? (Yes/No)	YES
9.2	Can the solution operate in a **fully air-gapped / offline** environment with **no internet access**? (Yes/No , if no, list every online dependency)	YES
9.3	Does any component (model, library, licence check, telemetry) require **outbound internet / callback** to any external server at install-time or run-time? (Yes/No , details)	NO. No application component, AI model, OCR engine, database, knowledge graph service, telemetry component, or runtime library requires outbound internet access or callback to any external server during normal operation. Any required software packages, models, and container images can be pre-packaged and installed through approved offline deployment procedures.
9.4	Does any component depend on a **public cloud API** (e.g., hosted LLM/OCR)? If yes, is a self-hosted / offline equivalent available?	NO
9.5	Do you collect any **telemetry / analytics**? Can it be fully disabled?	No mandatory telemetry is collected. System logging, audit trails, security logs, and operational monitoring are

		maintained locally for administration, troubleshooting, and compliance purposes. Optional analytics can be configured based on customer requirements and can be fully disabled if required. No usage data is transmitted outside the deployment environment.
9.6	List all **open-source libraries** and their licences (attach SBOM if available).	The proposed solution is primarily based on mature, widely adopted open-source technologies to ensure transparency, interoperability, long-term maintainability, and compliance with MDL's air-gapped deployment requirements. Equivalent alternatives are identified for flexibility, future upgrades, and supply-chain resilience. • Python 3.12 – Python Software Foundation (PSF) License Alternative(s): C++20, Rust • TypeScript 5.x – Apache License 2.0 Alternative(s): Java 21, Kotlin • React 19 – MIT License Alternative(s): Angular, Vue.js • FastAPI – MIT License Alternative(s): ASP.NET Core, Spring Boot • Uvicorn – BSD 3-Clause License Alternative(s): Hypercorn, Gunicorn • SQLAlchemy – MIT License Alternative(s): Prisma, Hibernate • Alembic – MIT License Alternative(s): Flyway, Liquibase • Pydantic – MIT License Alternative(s): attrs, Marshmallow • PyTorch 2.x – BSD-style License Alternative(s): TensorFlow, JAX • Scikit-learn – BSD 3-Clause License Alternative(s): XGBoost, CatBoost

		• OpenCV 4.x – Apache License 2.0 Alternative(s): OpenVINO, HALCON (Commercial) • Open3D – MIT License Alternative(s): PCL, PDAL, VTK • YOLOv11 – AGPL v3 Alternative(s): RT-DETR, Faster R-CNN, DETR • Segment Anything Model 2 (SAM 2) – Apache License 2.0 Alternative(s): Mask R-CNN, YOLOv11 Segmentation • Tesseract OCR – Apache License 2.0 Alternative(s): EasyOCR, Azure Read OCR (Commercial) • Hugging Face Transformers – Apache License 2.0 Alternative(s): vLLM, llama.cpp • Sentence Transformers – Apache License 2.0 Alternative(s): FlagEmbedding, Instructor Embeddings • all-mpnet-base-v2 Embedding Model – Apache License 2.0 Alternative(s): e5-large-v2, bge-large-en-v1.5 • spaCy – MIT License Alternative(s): Stanza, NLTK • ONNX Runtime – MIT License Alternative(s): OpenVINO Runtime, TensorRT • Llama 3.1 8B Instruct – Meta Llama 3.1 Community License Alternative(s): Mistral 7B, Phi-4, Gemma 3 • Qdrant Vector Database – Apache License 2.0 Alternative(s): FAISS, ChromaDB, Weaviate

		• ArcadeDB – Apache License 2.0 Alternative(s): Neo4j Community, JanusGraph, RDF4J • PostgreSQL – PostgreSQL License Alternative(s): MariaDB, Microsoft SQL Server • Redis – RSAL v2 / SSPL v1 Alternative(s): Valkey (BSD 3-Clause), RabbitMQ • Drools Rule Engine – Apache License 2.0 Alternative(s): OpenL Tablets, Easy Rules • Docker Engine – Apache License 2.0 Alternative(s): Podman • Kubernetes – Apache License 2.0 Alternative(s): K3s, OpenShift (Commercial)
9.7	**Country of origin** of every software/hardware/model component (cross-reference Section 8).	United States (USA) React 19, FastAPI, Uvicorn, SQLAlchemy, Alembic, Pydantic, PyTorch, OpenCV, Open3D, YOLOv11, Segment Anything Model 2 (SAM 2), Tesseract OCR, Hugging Face Transformers, ONNX Runtime, Llama 3.1 8B Instruct, PostgreSQL, Redis, Drools Rule Engine, Docker Engine, Kubernetes. United Kingdom (UK) all-mpnet-base-v2 Embedding Model, Sentence Transformers. France Scikit-learn. Germany spaCy. Italy ArcadeDB. Poland Qdrant. Netherlands Python 3.12.

		Microsoft (USA) TypeScript. Sweden (Alternative Components) Neo4j Community. United Kingdom / Germany (Alternative Components) e5-large-v2 Embedding Model (Microsoft Research UK), Instructor Embeddings (HKUST / USA Collaboration). United States / France (Alternative Components) vLLM, OpenVINO Runtime (Intel). United States / Canada (Alternative Components) TensorFlow, JAX. Japan (Alternative Components) CatBoost. Czech Republic (Alternative Components) EasyOCR. India (Applicable Standards / Deployment) CERT-In Cybersecurity Guidelines, Government of India Procurement & Data Sovereignty Requirements. Commercial Alternatives (Multiple Countries) HALCON (Germany), ASP.NET Core (.NET Foundation / USA), Spring Boot (VMware / USA), RabbitMQ (UK), Valkey (Linux Foundation / Global), MariaDB (Finland), Microsoft SQL Server (USA), OpenShift (USA).
9.8	Confirm compliance with the **Order (Public Procurement No. 1) dated** of the Department of Expenditure, Ministry of Finance, regarding restrictions on procurement from bidders / components sourced from countries sharing a land border with India and confirm registration with the Competent Authority where applicable. (Compliant / Not Compliant + details)	COMPLIANT.

9.9	Confirm that **no software, model, hardware or cloud dependency routes, stores or processes data in, or is controlled from, a country sharing a land border with India**. (Confirm / Explain)	CONFIRMED. No operational data, engineering data, project information, model inference requests, or system telemetry are routed to, stored in, processed by, or controlled from any country sharing a land border with India. All processing remains within the customer's deployment environment.
9.10	**Encryption at rest** , supported? Mechanism?	YES. Encryption at rest is supported. Data can be protected using AES-256 encryption through database-level encryption, disk-level encryption, encrypted storage volumes, and operating-system security controls. Encryption mechanisms can be configured according to MDL security policies and infrastructure standards.
9.11	**Encryption in transit** , supported? Mechanism?	YES. Encryption in transit is supported through TLS 1.3 and HTTPS. All communications between users, application services, databases, APIs, and internal components can be secured using industry-standard TLS encryption with customer-managed certificates.
9.12	**Authentication** , supported modes (local / Active Directory / LDAP)?	Supported authentication modes include: • Local user authentication • Microsoft Active Directory (AD) • LDAP integration • Single Sign-On (SSO) through enterprise identity providers • Role-based enterprise authentication policies Authentication methods can be configured according to MDL infrastructure requirements.
9.13	**Role-Based Access Control** , supported? (Yes/No)	YES

9.14	**Audit logging** of user and system actions , supported? (Yes/No)	YES
9.15	Compliance with applicable **Government of India / CERT-In cybersecurity guidelines**. (Details)	Designed to support compliance with applicable Government of India cybersecurity requirements and CERT-In guidelines through on-premise deployment, air-gapped operation, RBAC, audit logging, encryption at rest and in transit, secure authentication, and local hosting of all AI/ML components.

Fairness note: On-premises / air-gapped operability is a **genuine functional requirement** of a Defence Shipyard environment and is therefore an evaluated parameter, Inability to operate offline does not automatically disqualify a bidder but will be scored accordingly under Section 15 (Criterion 6), as it directly affects fitness for the intended environment.

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10. Functional Compliance Matrix

For **each** feature, indicate the delivery position honestly. Use **Yes / No** (or ✓ / ✗) in the boolean columns and give a short remark. This matrix is central to the evaluation.

Legend:

- **Expectation During MVP** , what you commit to demonstrate at the MVP/demonstration stage.
- **Expectation After Development** , the fully realised capability after the development phase.
- **Available in Current Product** , exists today, out-of-the-box.
- **Customizable with Current Product** , achievable by configuration/customisation of your existing product (minor/major).
- **New Development Required** , needs fresh development.

#	Feature / Requirement	Expectation During MVP	Expectation After Development	Available in Current Product	Customizable with Current Product	New Development Required	Remarks
1	Ingest 2D production drawings (PDF/TIFF/DWG/DXF)	YES	YES	NO	NO	YES	Document ingestion and parsing to be developed

2	Ingest 3D CAD models (STEP/IGES/IFC/native)	YES	YES	NO	NO	YES	Requires CAD integration layer
3	Ingest as-installed point cloud (E57/LAS/PTS)	NO	NO	NO	NO	NO	Can be added later as an addon as per MDL requirement
4	Ingest photographs / images of installation	NO	NO	NO	NO	NO	Can be added later as an addon as per MDL requirement
5	Register/align as-installed data to design coordinates	NO	NO	NO	NO	NO	Can be added later as an addon as per MDL requirement
6	Recognise affected systems	YES	YES	NO	NO	YES	Knowledge Graph Driven Impact analysis
7	Recognise **ventilation / HVAC duct** systems	NO	YES	NO	NO	YES	Domain-specific engineering modelling
8	Recognise **equipment / machinery**	NO	YES	NO	NO	YES	Entity extraction and dependency mapping
9	Detect **route change** (pipe/duct rerouting)	NO	YES	NO	NO	YES	Requires CAD topology analysis
10	Detect **equipment relocation**	YES	YES	NO	NO	YES	Dependency and

							location comparison
11	Detect **missing** components	NO	YES	NO	NO	YES	Requires CAD/as-built comparison
12	Detect **added / extra** components	NO	YES	NO	NO	YES	Requires CAD/as-built comparison
13	Quantify **dimensional / positional** deviation	NO	YES	NO	NO	YES	Point cloud/CAD comparison module
14	Classify deviation **severity** against tolerances	YES	YES	NO	NO	YES	Constraint Validation Engine
15	**Confidence score**	YES	YES	NO	NO	YES	Risk Scoring Engine
16	**Visual overlay** of the system	NO	YES	NO	NO	YES	Dashboard visualisation capability
17	Automatic **report** generation	YES	YES	NO	NO	YES	Impact assessment reports
18	**Export / hand-off** to design department	YES	YES	NO	NO	YES	PDF/export package generation
19	**Revision / version** comparison of drawings	NO	YES	NO	NO	YES	Design baseline comparison
20	**Review & approval** workflow	YES	YES	NO	NO	YES	Engineering change approval workflow
21	**Role-based** user & access management	YES	YES	NO	NO	YES	RBAC implementation

22	**Audit trail**	YES	YES	NO	NO	YES	Complete activity traceability
23	**Dashboard / analytics**	YES	YES	NO	NO	YES	Unified management dashboard
24	Export to **PDF** / standard formats	YES	YES	NO	NO	YES	Standard reporting exports
25	**REST API** for integration	YES	YES	NO	NO	YES	Integration architecture
26	**Offline / air-gapped** operation	YES	YES	NO	NO	YES	Designed for on-prem deployment

Add rows for any additional capability you consider material.

- * IF any field is not applicable, please indicate “Not Applicable (N/A)” in relevant field.

11. Hardware & Infrastructure Requirements

State the on-premise hardware your solution needs at each tier. This must be realistic and match the models disclosed in Section 8.

Resource	Minimum (Dev / Test Workstation)	Recommended (This Proposal - Budgeted, ~10,000-node scale)	Enterprise (Full-Ship Scale - Future Production Rollout)
CPU	8-core (Intel i7 / AMD Ryzen 7), 32GB RAM	Dual Intel Xeon Silver 4310 (24 cores) or AMD EPYC 7313, 128GB RAM per node x2 nodes	Dual Intel Xeon Gold / AMD EPYC (64+ cores), 256GB+ RAM, clustered x4-6 nodes
GPU (model & VRAM)	NVIDIA RTX 3060 12GB (CAD viz only, no ML training)	NVIDIA RTX 3090 or equivalent, 24GB VRAM x1-2 (PyTorch surrogate + XGBoost training); CPU-only inference fallback	NVIDIA RTX A6000 / A100 40-80GB VRAM x2-4, clustered for multi-ship-class retraining
RAM	32 GB	128 GB per server node (256GB aggregate)	512GB - 1TB aggregate across cluster
Storage (type & capacity)	1TB NVMe SSD (local)	4TB NVMe (hot, DB/graph) + 20TB RAID6 NAS/MinIO object store (CAD, sim files, model artefacts)	10TB+ NVMe hot tier + 100TB+ scalable object storage, replicated

Operating System	Ubuntu Desktop LTS 22.04/24.04 (Open Source)	Ubuntu Server LTS 24.04 (Open Source, GPL) - containerized via Docker-Compose	Ubuntu Server LTS 24.04 / RHEL, Kubernetes-orchestrated, HA cluster
Database	PostgreSQL 15 (single instance, containerized)	PostgreSQL 15 + Neo4j (Dependency Knowledge Graph) + Redis 7.x + MinIO, all containerized on-premise	PostgreSQL 15 (HA/replicated) + Neo4j Enterprise (clustered) + Redis Sentinel + MinIO distributed
Networking & Security	Standard LAN, local firewall	Managed switch, dedicated air-gap boundary/firewall appliance, mTLS between microservices	Zero-trust network fabric, redundant switching, IDS/IPS, SOC integration
Any specialised hardware (scanner, edge device)	Not required	10 KVA online UPS; backup/tape storage unit for offline update packages; biometric/smart-card RBAC reader (optional)	N+1 UPS/power redundancy; automated offline patch-delivery appliance; edge scanning stations at drawing office
Estimated indicative hardware cost band	Rs. 1.5 - 2.5 Lakh (per workstation)	Rs. 26.8 Lakh (full budgeted procurement - see Sheet 3 for line-item detail)	Rs. 80 Lakh - 1.5 Crore+ (multi-node HA cluster, full-ship scale; outside current PS-71 budget)

12. Product Maturity

#	Question	Response
12.1	Best description of the offering: Deployed Product ☐ / MVP ☐ / POC ☐ / Prototype ☐ / Research ☐ / Custom Development ☐	Research
12.2	Number of **production installations** of the core product	0
12.3	Has it been **deployed elsewhere** for a comparable use-case? (Where?)	No
12.4	Product version / release history (brief)	Not Applicable
12.5	What % of this challenge's needs are met **today** vs need **new development**? (cross-ref Section 10)	Core architectural components, workflow management, dashboarding, RBAC, audit logging, document processing, knowledge-graph infrastructure, and AI modules are available within existing Arcaisys platforms. Shipyard-specific functionality requires development and integration.

13. Capability Demonstration (Small, Time-Boxed)

Intent (fairness): This is an **existing-capability demonstration**, not a mandate to build a bespoke solution free of charge. You may demonstrate using **(a)** a small representative sample dataset provided by the shipyard, **or (b)** your own existing product/prior implementation on comparable data. The objective is to assess technical maturity and directional fit , not completeness of a customised solution.

Suggested minimal scope (indicative , demonstrate any subset you can credibly show):

1. Demonstrate the key AI/GenAI capabilities relevant to the proposed solution, including the approach workflow and expected outcomes.
2. Demonstration of handling relevant data inputs, processing methods and generation of AI-driven outputs, insights or recommendations.
3. Demonstrate the user interactions, reporting, integration or operational capability of the system
4. Show that the workflow can run **without internet** (or explain the offline path).

#	Question	Response
13.1	Can you demonstrate the above on the shipyard's sample data? (Yes/No)	YES
13.2	Alternatively, can you demonstrate equivalent capability on your **own existing** data/product? (Yes/No)	NO
13.3	Mode of demonstration (Live on-site ☐ / Live remote ☐ / Recorded video + walkthrough ☐)	Live On-Site
13.4	Indicative lead time needed to prepare the demonstration	2-4 weeks from receipt of representative sample data and demonstration scope.
13.5	Any data/support required from the shipyard for the demonstration	The proposed demonstration can be executed using a limited, sanitized dataset provided by the shipyard. The following support would be required: • 5–10 representative engineering drawings (PDF/DWG/DXF) covering one or two ship systems. • Corresponding 3D CAD models (STEP/IGES/IFC or native CAD format) for the same assemblies. • Associated Bill of Materials (BOM) in Excel/CSV/XML format. • A small sample (10–20 documents) of applicable classification rules, standards, or internal design guidelines.

		• One or two representative Engineering Change Requests (ECR/ECO) or design revision examples illustrating a typical design modification. • Basic metadata linking drawings, CAD models, and BOM (e.g., drawing number, revision, part number). No proprietary production environment, source code, or live operational systems are required for the demonstration. All processing will be performed on an isolated, air-gapped demonstration environment.

14. Risks, Assumptions & Dependencies

1. List your **top implementation risks** for this challenge and the **mitigation** for each.
2. List key **assumptions** underlying your solution.
3. List **dependencies** on the shipyard (data, access, scanning, infrastructure, SMEs).

#	Risk	Likelihood (H/M/L)	Impact (H/M/L)	Mitigation
1	Incomplete, inconsistent or poor-quality design, BOM and rule data affecting AI accuracy	M	H	Implement automated data validation, schema checks, confidence scoring, and SME-assisted data cleansing before model training. Support incremental onboarding of data sources.
2	Integration complexity with existing CAD, PLM and ERP systems	M	H	Use modular adapter-based architecture with standard APIs and staged integration. Begin with file-based import/export during MVP and progressively enable live integrations.
3	Frequent revisions to class rules, standards and regulations causing outdated recommendations	M	H	Maintain a version-controlled knowledge base with automated document ingestion, rule versioning, configurable update workflows and periodic validation by domain experts.

2. Key Assumptions

- MDL will provide representative design data, 3D CAD models, BOMs, historical revisions and applicable class-rule documents for the pilot phase.
- Necessary software licenses for class rules, standards and engineering documents will be available for system ingestion.
- Subject Matter Experts (SMEs) from design, production and configuration management will be available for requirement validation, rule interpretation and acceptance testing.
- Required CAD, PLM and ERP interfaces or data export mechanisms will be accessible during integration.
- Deployment infrastructure meeting the agreed hardware specifications (CPU, GPU, storage and networking) will be available within MDL's secure on-premise environment.

- The solution will operate entirely within MDL's air-gapped infrastructure with no dependency on external cloud services.
- Acceptance criteria, accuracy targets and pilot scope will be jointly finalized before system validation.

3. Dependencies on the Shipyard

Category	Dependency
Engineering Data	Access to representative CAD models (e.g., STEP, IGES, native CAD), BOMs, engineering drawings, metadata, revision history and configuration baselines.
Reference Documents	Licensed class rules, standards, regulations, design manuals and engineering procedures in digital or scanned formats.
System Access	Read-only or controlled access to PLM, ERP, CAD and document management systems, or periodic data exports where direct integration is unavailable.
Infrastructure	On-premise servers, GPU resources (where required), secure storage, networking, authentication services (AD/LDAP), backup facilities and cybersecurity controls.
Subject Matter Experts	Design engineers, configuration management teams, classification and regulatory specialists, production engineers and QA personnel for validation, feedback and user acceptance testing.
Security & Compliance	Access approvals, cybersecurity clearance, deployment permissions and compliance with MDL/GoI security policies for operation in an air-gapped environment.

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15. Evaluation Criteria & Scoring Methodology (Published for Transparency)

15.1 Weightage

Responses are scored out of **100** across eight criteria. The same rubric is applied uniformly to all bidders by an evaluation committee.

#	Criterion	Weight
1	Relevant Experience (AI/GenAI in engineering, manufacturing, shipbuilding, defence, govt, PSU, technical KM)	20
2	Understanding of the Problem Statement	10
3	Solution Architecture	15
4	AI/GenAI Technical Capability	15

5	Similar Project Experience (drawing interpretation / CV inspection / deviation / as-built vs as-designed)	15
6	Security & Air-Gapped Readiness	10
7	Product Maturity / Feature Compliance	10
8	Team Capability	5
	Total	**100**

15.2 Scoring Rubric (Sub-Parameters)

Each criterion is scored from its sub-parameters. Every claim must be evidenced; **unevidenced claims score at the lowest band.**

Criterion 1 , Relevant Experience (20)

Sub-parameter	**Max**
Number of evidenced AI/GenAI projects in the preference domains	8
Depth of domain relevance (shipbuilding/defence/PSU rated highest)	7
Experience deploying in secure / on-prem / air-gapped environments	5

Criterion 2 , Understanding of Problem Statement (10)

Sub-parameter	**Max**
Clarity of problem articulation & identification of real pain / edge cases (occlusion, intentional vs defect, revisions)	6
Quality of clarification questions, assumptions & dependencies	4

Criterion 3 , Solution Architecture (15)

Sub-parameter	**Max**
Completeness & soundness of end-to-end architecture	6
Data-flow clarity & handling of 2D / 3D / point-cloud / image inputs	4
Integration with CAD/PLM & as-fitted hand-off	3
Scalability to full-ship / large volumes	2

Criterion 4 , AI/GenAI Technical Capability (15)

Sub-parameter	**Max**
Appropriateness & soundness of CV/3D/OCR/RAG/LLM choices for the task	6

False-positive reduction, confidence scoring, explainability	4
Human-in-the-loop, continuous learning, accuracy measurement	3
Completeness & credibility of AI model disclosure (Section 8)	2

Criterion 5 , Similar Project Experience (15)

Sub-parameter	**Max**
Prior projects specifically on deviation / as-built vs as designed / drawing interpretation / 3D comparison	9
Strength of evidence (live demo / video / client reference / certificate)	6

Criterion 6 , Security & Air-Gapped Readiness (10)

Sub-parameter	**Max**
On-prem / air-gapped / offline operability (evidenced)	4
Country-of-origin disclosure & compliance with land-border-country procurement restrictions	3
Encryption, authentication, RBAC, audit, cybersecurity certifications	3

Criterion 7 , Product Maturity / Feature Compliance (10)

Sub-parameter	**Max**
Maturity stage (Deployed Product > MVP > POC > Prototype > Research)	4
Functional Compliance Matrix , proportion available/configurable vs new development	4
Number of production installations	2

Criterion 8 , Team Capability (5)

Sub-parameter	**Max**
Relevant multidisciplinary team (AI, CV, 3D, CAD, mechanical/marine, shipbuilding SME)	3
Certifications (ISO/CMMI/security) & delivery track record	2

16. Submission Checklist

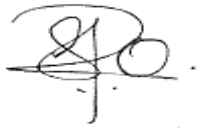
- ☑ Section 3 , Bidder information & team composition completed
- ☑ Section 4 , Relevant & similar experience with evidence attached (Work Orders / Completion Certificates / references)
- ☑ Sections 5–6 , Technical Solution Summary (≤ 10 pages) + architecture & data-flow diagrams

- ☑ Section 7 , AI/GenAI capability responses
- ☑ Section 8 , Technology stack & AI model disclosure tables completed
- ☑ Section 9 , Security & air-gapped questionnaire completed
- ☑ Section 10 , Functional Compliance Matrix completed
- ☑ Section 11 , Hardware requirements
- ☑ Section 12 , Product maturity
- ☑ Section 13 , Capability demonstration readiness
- ☑ Section 14 , Risks, assumptions, dependencies
- ☑ Annexure A , Signed Declaration of Correctness

Annexure A , Declaration of Correctness

We, **Arcaisys Private Limited**, hereby declare that:

1. All information furnished in this questionnaire is **true, correct and complete** to the best of our knowledge, and is supported by the evidence attached.
2. We understand that any **misrepresentation or suppression** of material facts may lead to disqualification at any stage.
3. The AI models, software, hardware and cloud components disclosed in Sections 8–9 are **complete and accurate**, and we comply with applicable **Government of India procurement and cybersecurity guidelines**, including restrictions on procurement from countries sharing a land border with India.
4. We understand this stage is a **capability and compliance assessment**, and that any demonstration may be based on our existing products or representative data.

Authorised Signatory: 

Name & Designation: Mr. Leejo Xavier Puravathan, Director

Date & Place: 17th July 2026, Mumbai

Company Seal: 